Hints: Trigonometry – 2. Roots of unity and the complex exponential

Class work

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. For the filter: if $n \nmid m$ then $\varepsilon^m \neq 1$ and $\sum_{k=0}^{n-1}(\varepsilon^m)^k$ is a geometric sum equal to $\frac{1-\varepsilon^{mn}}{1-\varepsilon^m} = 0$; if $n \mid m$ every term is 1. For the subsets, encode them by the generating function $f(x) = \sum_s a_s x^s$, where a_s is the number of subsets with sum s; show that $f(x) = \prod_{k=1}^{n}(1+x^k)$.

Hint 2. With $\omega = e^{2\pi i/3}$ the filter gives $\sum_{3|s} a_s = \frac{1}{3}(f(1) + f(\omega) + f(\omega^2))$. To compute $f(\omega) = \prod_k(1+\omega^k)$, group the factors in blocks of three consecutive k: $(1+\omega)(1+\omega^2) = 1$, so a full block contributes 2.

Hint 3. Writing $n = 3q + r$ one gets $f(\omega) = f(\omega^2) = 2^q$ for $r = 0, 2$ and $f(\omega) + f(\omega^2) = 2^q$ for $r = 1$; so the number of subsets (the empty one included) is $\frac{2^n+2^{q+1}}{3}$ if $3 \nmid n-1$ and $\frac{2^n+2^q}{3}$ if $n \equiv 1 \pmod 3$.

Hint 4. The same filter applied to $g(x) = (1+x)^n = \sum_k \binom{n}{k}x^k$ gives $\sum_{3|k}\binom{n}{k} = \frac{1}{3}(2^n + (1+\omega)^n + (1+\omega^2)^n)$; since $1+\omega = e^{i\pi/3}$, the answer is $\frac{2^n+2\cos(\pi n/3)}{3}$.

Problem 2

Hint 1. Divide $z^n - 1 = \prod_{k=0}^{n-1}(z-\varepsilon^k)$ by $z-1$: $1 + z + \cdots + z^{n-1} = \prod_{k=1}^{n-1}(z-\varepsilon^k)$. Put $z = 1$, take moduli, and use $|1 - e^{i\alpha}| = 2\sin\frac{\alpha}{2}$ for $0 < \alpha < 2\pi$.

Hint 2. For the sum, rewrite one term algebraically: for $|w| = 1$, $w \neq 1$, the number $u = \frac{1}{1-w}$ satisfies $\bar{u} = \frac{1}{1-1/w} = \frac{w}{w-1} = 1 - u$, hence $\frac{1}{|1-w|^2} = u\bar{u} = u - u^2$. So you need $\sum_k \frac{1}{1-\varepsilon^k}$ and $\sum_k \frac{1}{(1-\varepsilon^k)^2}$.

Hint 3. With $f(z) = \frac{z^n-1}{z-1} = \prod_{k=1}^{n-1}(z-\varepsilon^k)$ these sums are $(\log f)'(1)$ and $-(\log f)''(1)$. From $f(1) = n$, $f'(1) = \frac{n(n-1)}{2}$, $f''(1) = \frac{n(n-1)(n-2)}{3}$ one gets $\frac{n-1}{2}$ and $-\frac{(n-1)(n-5)}{12}$; add them.

Problem 3

Hint 1. Write $\varepsilon = e^{2\pi ij/n}$ and $s = \varepsilon + \varepsilon^{-1} = 2\cos\frac{2\pi j}{n}$, a real number. Then $\varepsilon^2 = s\varepsilon - 1$: use this relation to reduce powers of ε.

Hint 2. Prove by induction that $\varepsilon^k = A_k(s)\varepsilon + B_k(s)$ with $A_k, B_k \in \mathbb{Z}[x]$ (take $A_{k+1} = xA_k + B_k$, $B_{k+1} = -A_k$). Hence $p(\varepsilon) = A(s)\varepsilon + B(s)$ with $A, B \in \mathbb{Z}[x]$.

Hint 3. If ε is not real, compare imaginary parts: $A(s)\operatorname{Im}\varepsilon = 0$ forces $A(s) = 0$ and $z = B(s)$. If $\varepsilon = \pm 1$, $z = p(\pm 1) \in \mathbb{Z}$ is a constant polynomial.

Hint 4. Finally, $s = 2\cos\frac{2\pi j}{n} = D_j(2\cos\frac{2\pi}{n})$, where $D_0 = 2$, $D_1 = x$, $D_{k+1} = xD_k - D_{k-1}$ are integer polynomials with $D_k(2\cos\theta) = 2\cos k\theta$. Take $q = B \circ D_j$.

Problem 4

Hint 1. Work in $\mathbb{C}$ and follow the hint of the sheet: write the vertices as a vector $Z = (z_0, \ldots, z_{n-1})$ and express the apex of the isosceles triangle erected on $z_j z_{j+1}$ with apex angle

φ in terms of z_j, z_{j+1}; orient the polygon counterclockwise, so that "outwards" means to the right of the directed side $z_j \to z_{j+1}$. (Check that, since n is odd, no angle equals π.)

Hint 2. The apex is $w_j = \frac{1+ic_k}{2} z_j + \frac{1-ic_k}{2} z_{j+1}$ with $c_k = \cot \frac{\pi k}{n}$ for $\varphi = \frac{2\pi k}{n}$; the same formula covers the inward case $\frac{2\pi k}{n} > \pi$ (then $c_k < 0$). So the step is the linear map $A_k = \frac{1+ic_k}{2} E + \frac{1-ic_k}{2} S$, where S is the cyclic shift, and the $(n-1)$-st polygon is $A_{n-2} \cdots A_1 Z$.

Hint 3. All A_k are diagonal in the basis $v_m = (1, \varepsilon^m, \ldots, \varepsilon^{(n-1)m})$, $m = 0, \ldots, n-1$ (eigenvectors of S). Compute the eigenvalue $\lambda_{k,m} = \frac{1+ic_k}{2} + \frac{1-ic_k}{2} \varepsilon^m$. Since $1 \pm ic_k = \pm \frac{i e^{\mp i\pi k/n}}{\sin(\pi k/n)}$, one has $1 - ic_k = -(1+ic_k)\varepsilon^k$, so $\lambda_{k,m} = \frac{1+ic_k}{2}(1 - \varepsilon^{k+m})$: it vanishes exactly when $m \equiv -k \pmod n$.

Hint 4. As k runs over $1, \ldots, n-2$, $-k$ runs over all residues $2, \ldots, n-1$; so in the expansion $Z = \sum a_m v_m$ every component with $m \geqslant 2$ is killed, while those with $m = 0, 1$ survive. Hence the final vertices are $a_0 + b\,\varepsilon^j$: a regular n-gon centred at the centroid.

Problem 5

Hint 1. Since ε is a primitive m-th root of unity, $\prod_{k=0}^{m-1}(x - \varepsilon^k) = x^m - 1$; read off from Vieta the elementary symmetric functions σ_j of $1, \varepsilon, \ldots, \varepsilon^{m-1}$: $\sigma_1 = \cdots = \sigma_{m-1} = 0$, $\sigma_m = (-1)^{m-1}$.

Hint 2. Lower bound: rank is subadditive, and $\sum_k (E + \varepsilon^k A) = mE + \sigma_1 A = mE$.

Hint 3. Upper bound: the factors $E + \varepsilon^k A$ commute, so $\prod_k (E + \varepsilon^k A) = E + \sigma_1 A + \cdots + \sigma_m A^m = E + (-1)^{m-1} A^m = 0$. Apply Sylvester's inequality $\operatorname{rank}(XY) \geqslant \operatorname{rank} X + \operatorname{rank} Y - n$ repeatedly to this product of m factors.